SHELFWISE — a library system that moves the queue

Demand-aware, race-free, tamper-evident library management in Java

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01 Problem

Typical library software copies the paper register: one fixed loan period for every book, a search that fails on one typo, an issue button that can lend the last copy twice, and a fines history anyone with database access can quietly edit.

02 Objectives

- OOP design: abstract Member → Student / Faculty, LoanPolicy interface, records, exceptions
- JDBC + SQLite with one transaction per operation
- Loans that shorten only when members are waiting
- Race-free issue; SHA-256 hash-chained audit ledger
- Typo-tolerant search; ISBN checksum repair

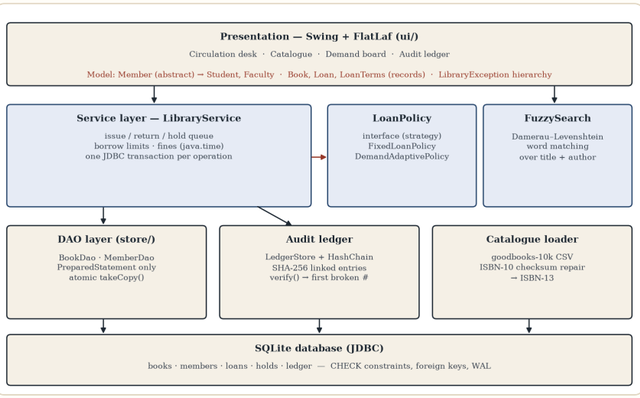
03 Data finding

6,587 of 10,000 ISBNs had lost their leading zeros

1,000 titles · 1,250 copies kept

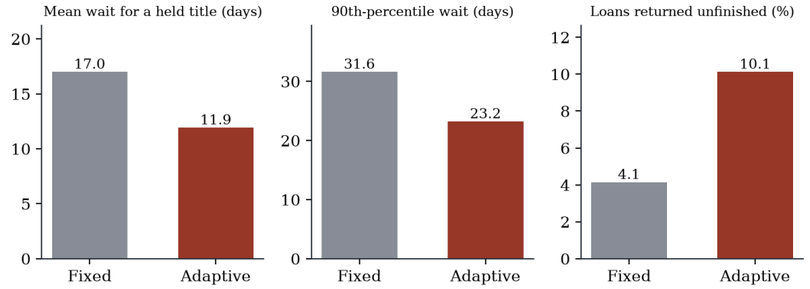
99.8% of repairs match the dataset's own ISBN-13

04 Architecture



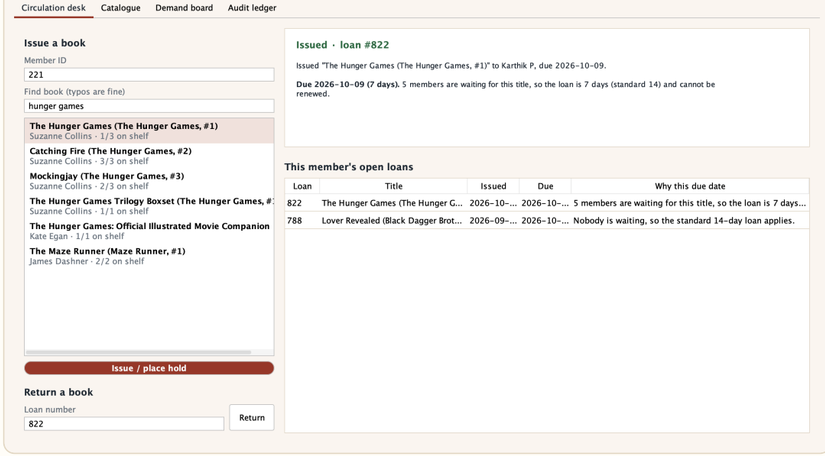
05 Demand-adaptive loans

```
days = max(7, round(base / (1 + 0.5 × members waiting))) · nobody waiting → standard 14 / 28 days
```



Simulated year, 400 members, demand from real Goodreads popularity, 5 seeds. Cost reported honestly: more loans returned unfinished.

06 The application



07 Results

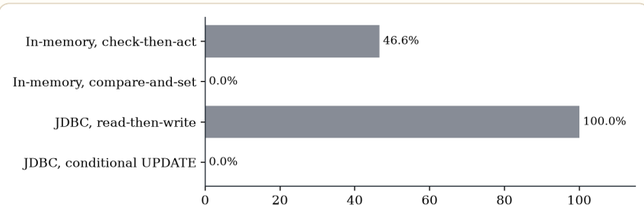
−29.9% wait for a held title (17.0 → 11.9 days)

0 / 150 double issues (naive: 150 / 150)

1,000 / 1,000 tamper attempts caught and located

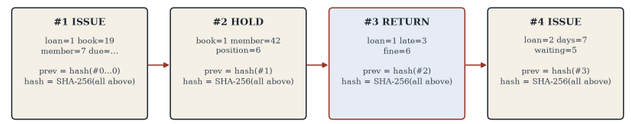
Search with one typo: 75.6% vs 1.6% · 39/39 JUnit tests pass · 74.5% core line coverage

08 Race-free issue




```
UPDATE books SET available = available - 1 WHERE id = ? AND available > 0
```

09 Tamper-evident ledger



Edit any field of #3 and its hash no longer matches; delete or reorder it and #4's prev link breaks.

10 Conclusion & future scope

Popular books reach readers sooner, the system stays correct under concurrent use, and its history is provably intact. Next: department pilot, off-site hash anchoring, SMS hold alerts.